

Predicting outcomes from the topology of behavior in brain lesion patients

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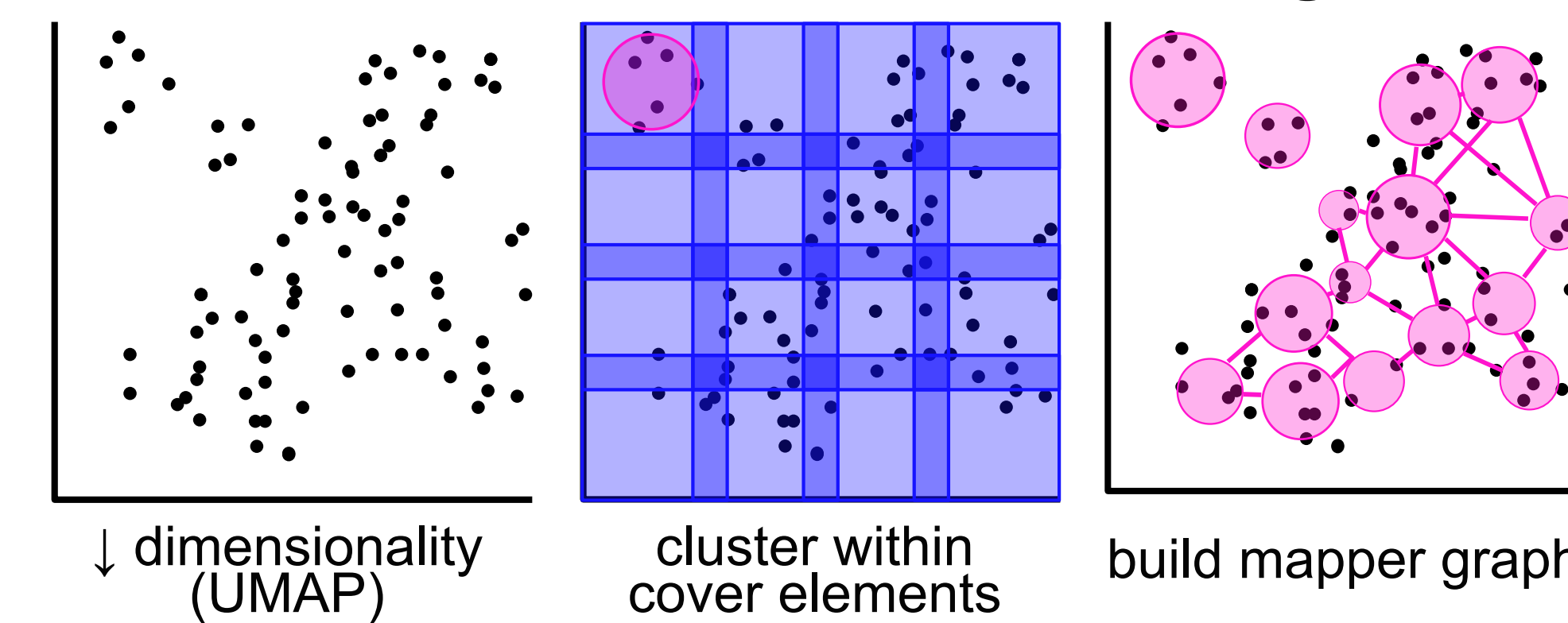
Introduction

- Brain lesion patients show highly-variable and complex patterns of behavioral impairments.
- Data-driven predictions of outcome and/or lesion location from behavior are hindered by the "curse of dimensionality".¹
- Topological Data Analysis (TDA) reveals nonlinear structure in high-dimensional data that standard approaches (e.g., PCA, clustering) may miss.^{2,3}

Goal: Visualize the shape of the distribution of behavioral deficits in brain lesion patients, characterize underlying brain damage, and predict functional recovery

Methods

TDA-Mapper Algorithm



WashU Cohort (n=85)

TDA-Mapper >> acute behavior

NEUROANATOMY

Lesion mapping (proportional subtraction) to identify brain lesion locations in a node

OUTCOME PREDICTION

Acute Mapper graph used to predict 3-month functional outcomes (LOO ridge reg.)

54.5 ± 10.6 years old at time of stroke, 52% women, 13.6 ± 2.63 years education
94% right-handed
64.7% Black/Afr-Am
98% non-Hispanic

Iowa Cohort (n=255)

TDA-Mapper >> chronic behavior

NEUROANATOMY

Lesion mapping (proportional subtraction) to identify brain lesion locations in a node

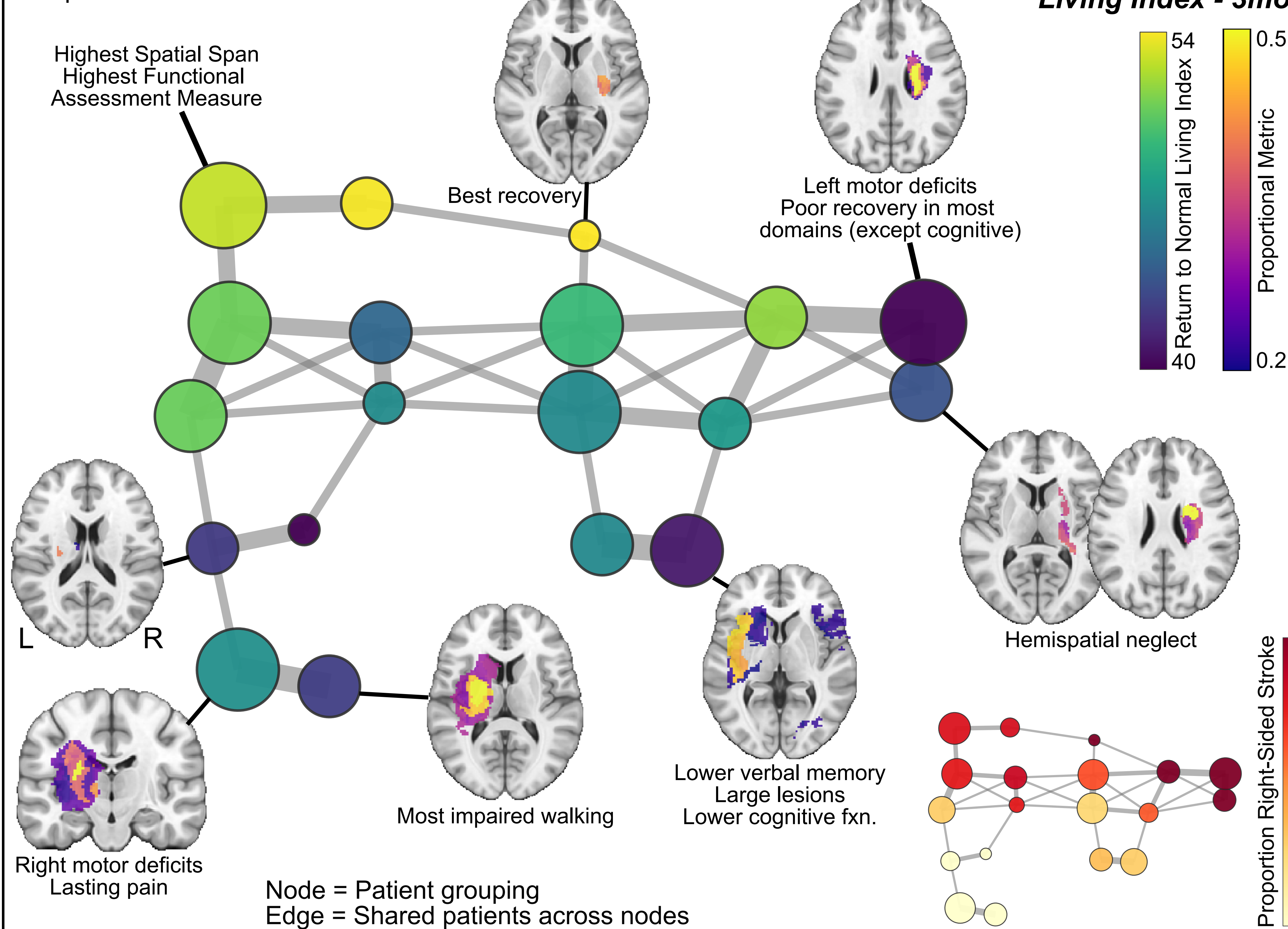
OUTCOME PREDICTION

Cross-cohort prediction of behavior and lesion location (future direction)

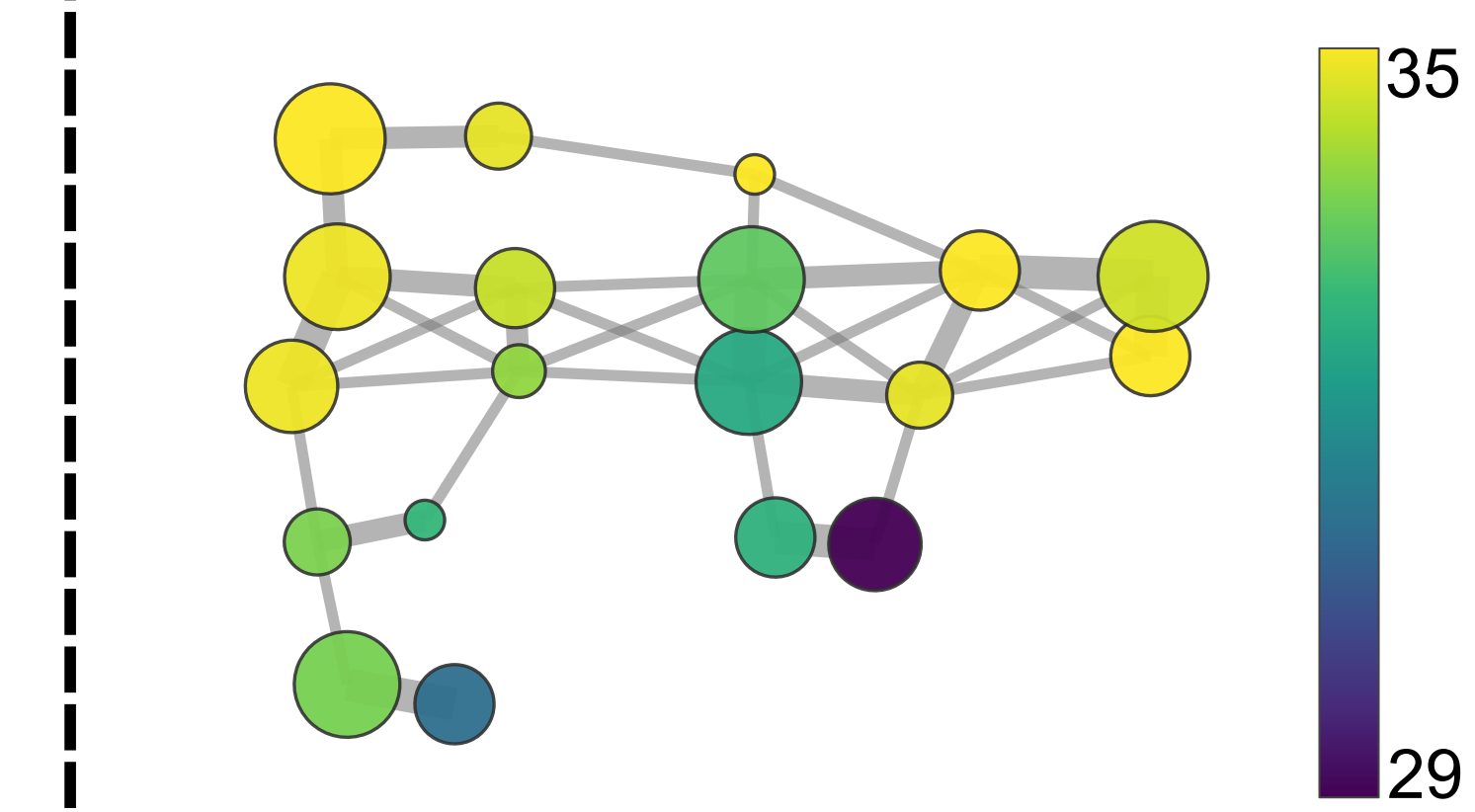
52.1 ± 13.9 years old at time of lesion, 50% women, 13.6 ± 2.80 years education
88% right-handed
97.0% White
96% non-Hispanic

WashU Cohort - Unilateral Stroke

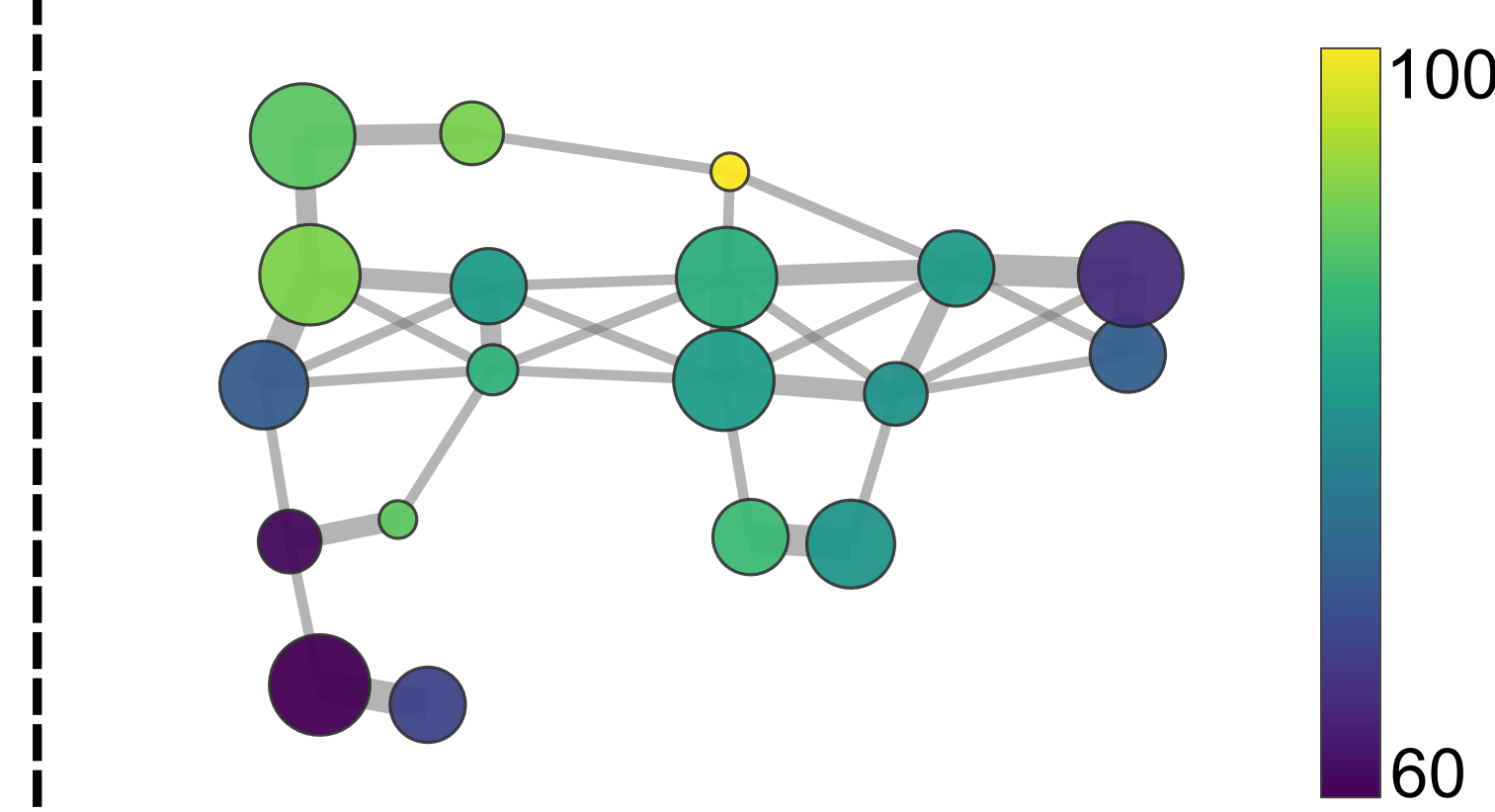
Graph constructed from acute behavior



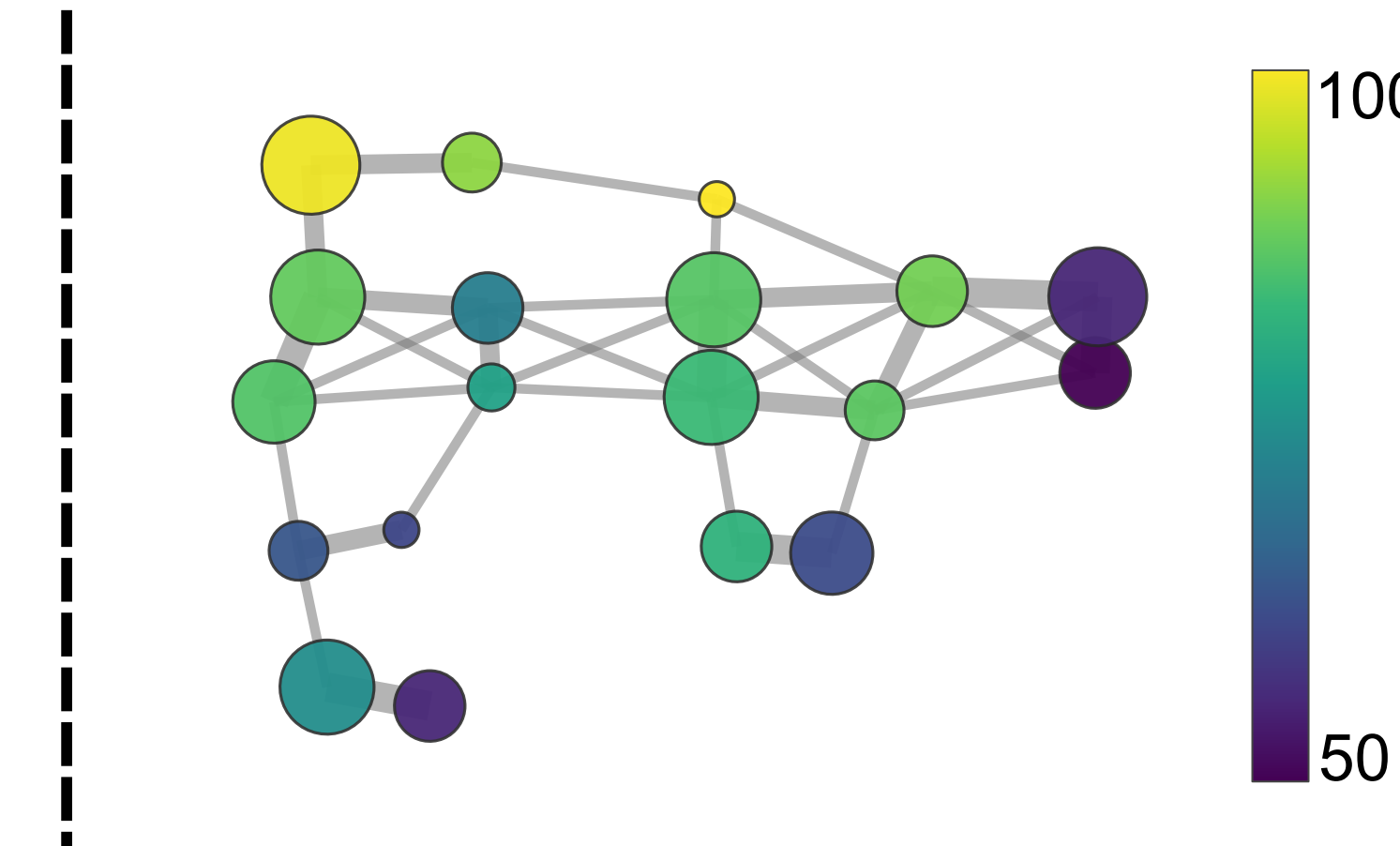
Fxl. Indep. (Cognitive) - 3mo



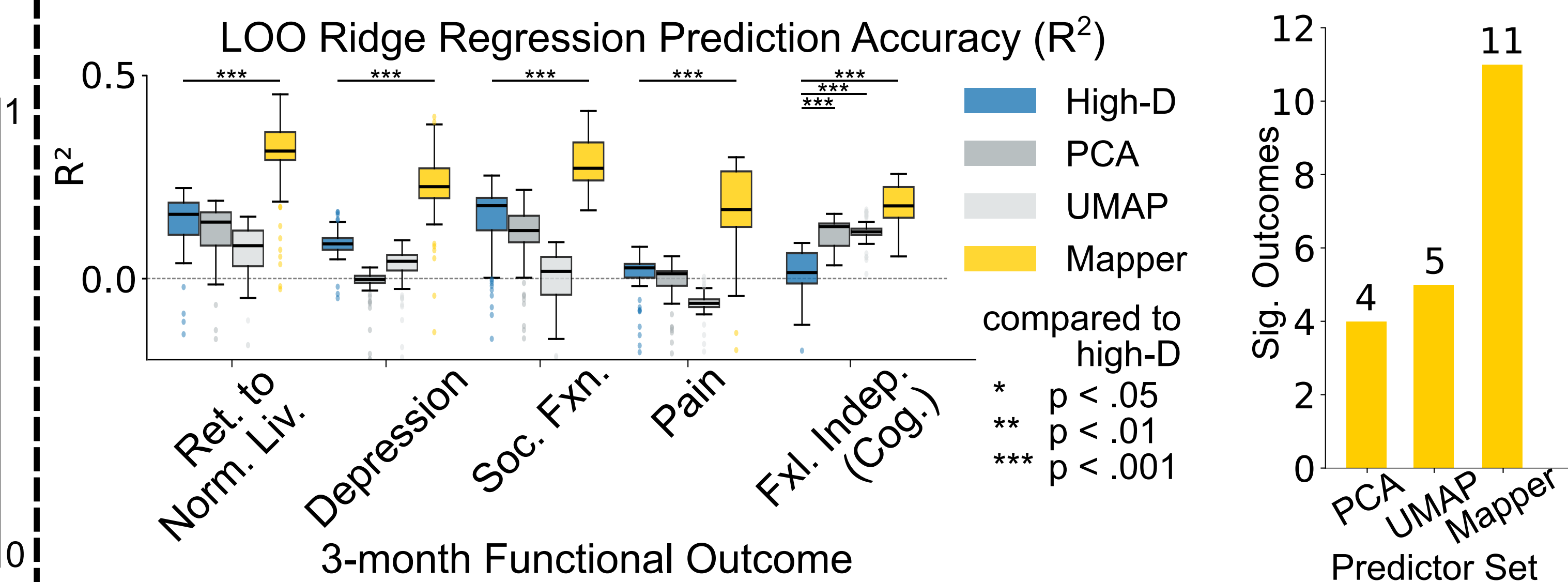
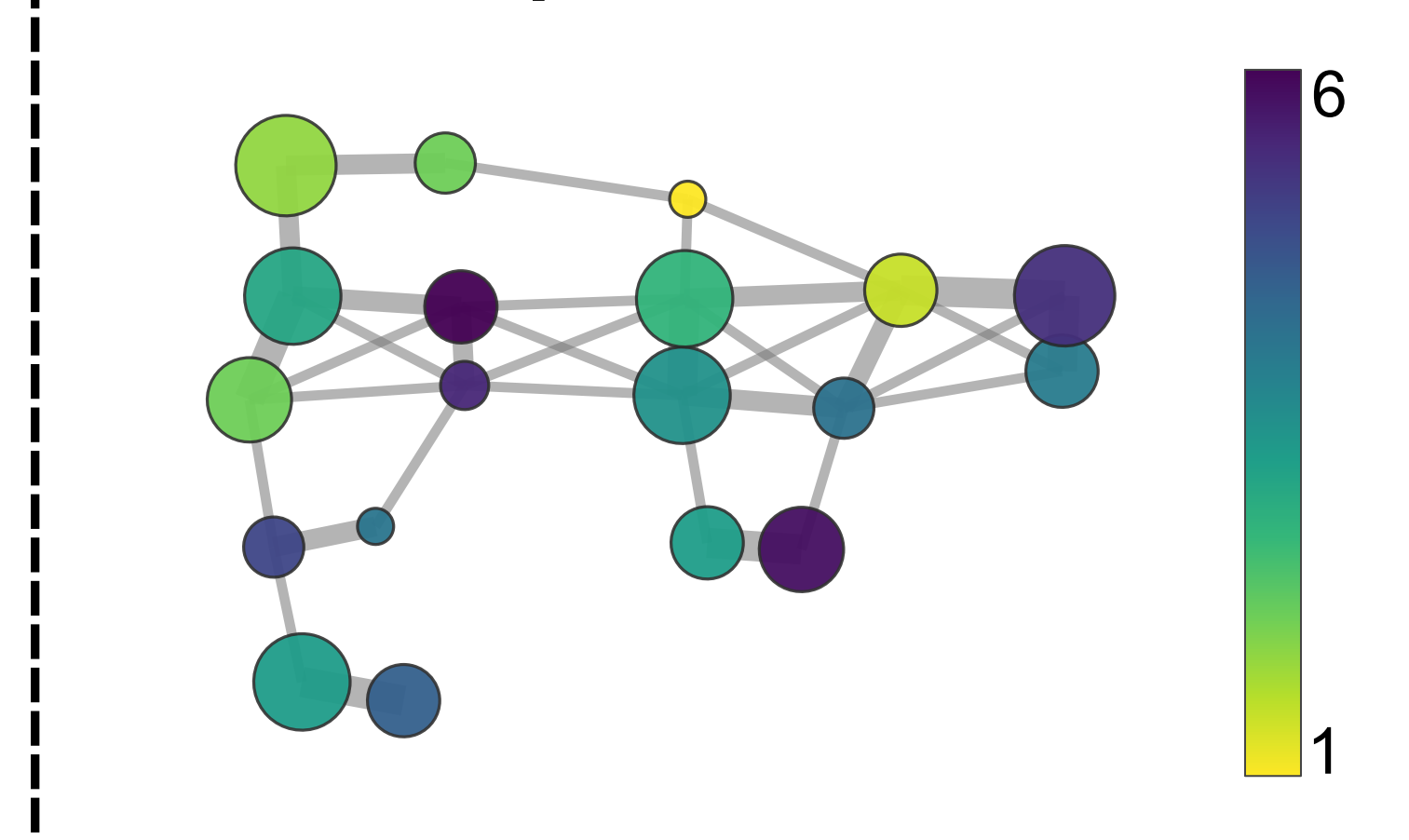
Pain - 3mo



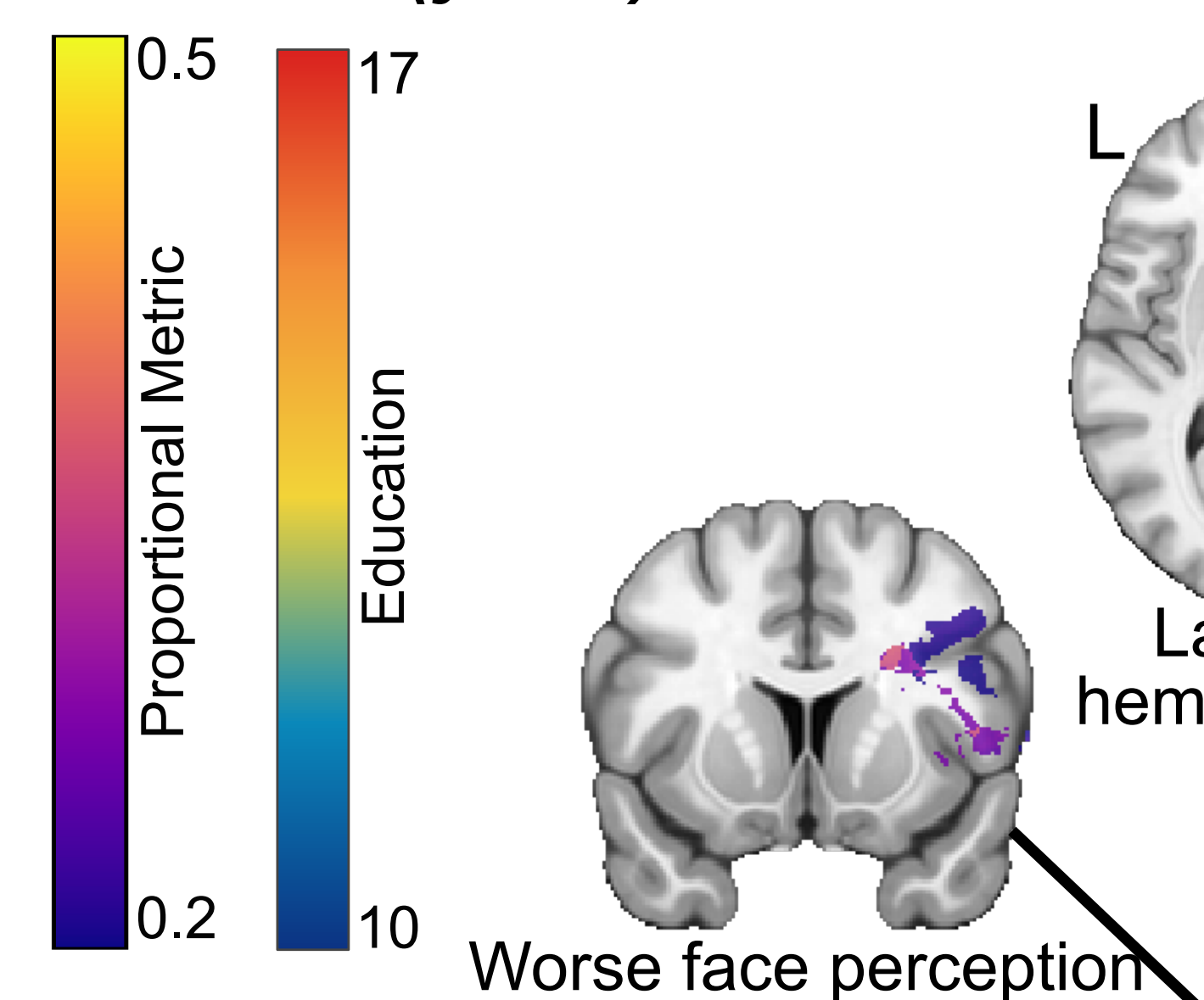
Social Fxn. - 3mo



Geriatric Depression Scale - 3mo

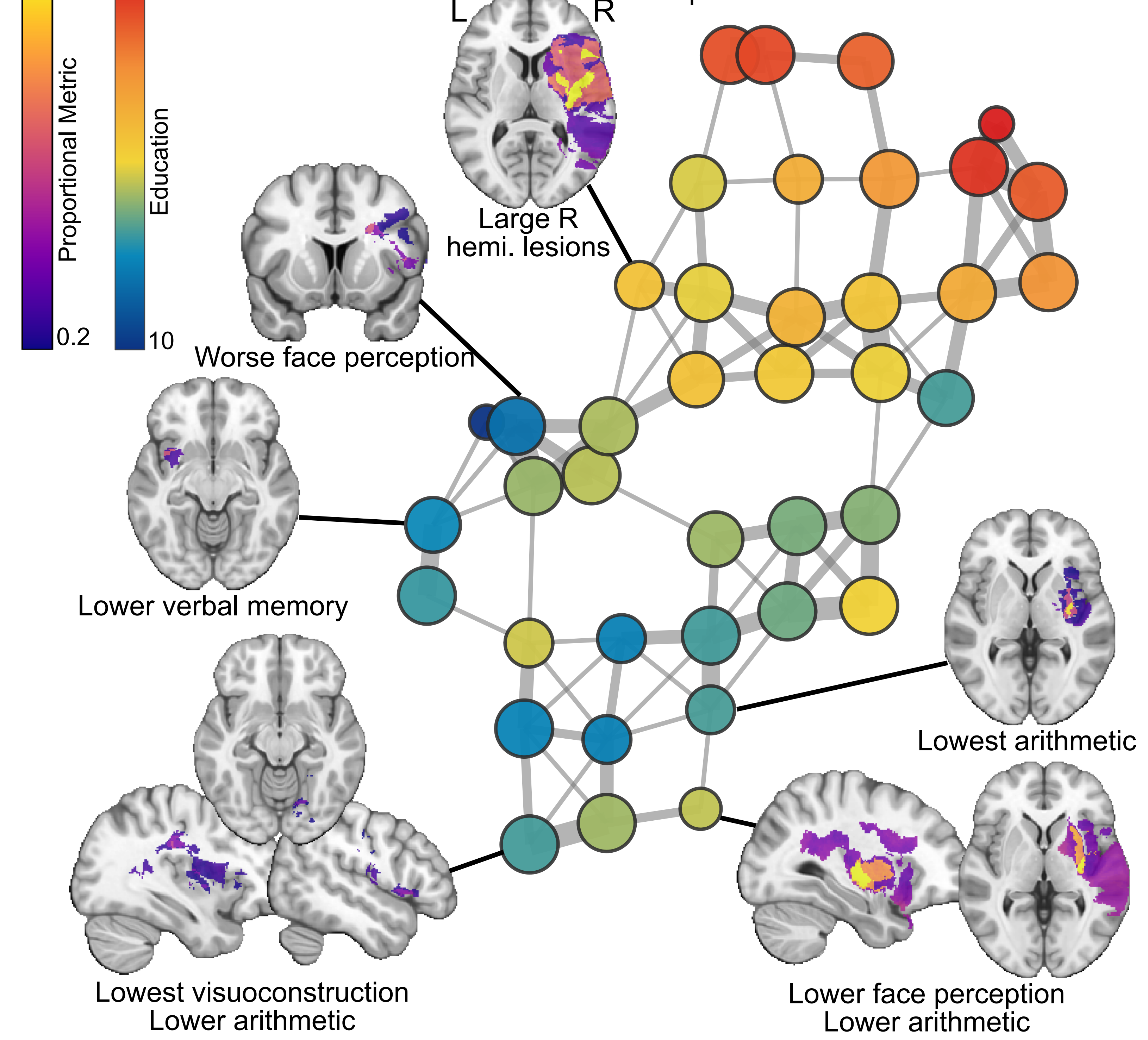


Education (years)

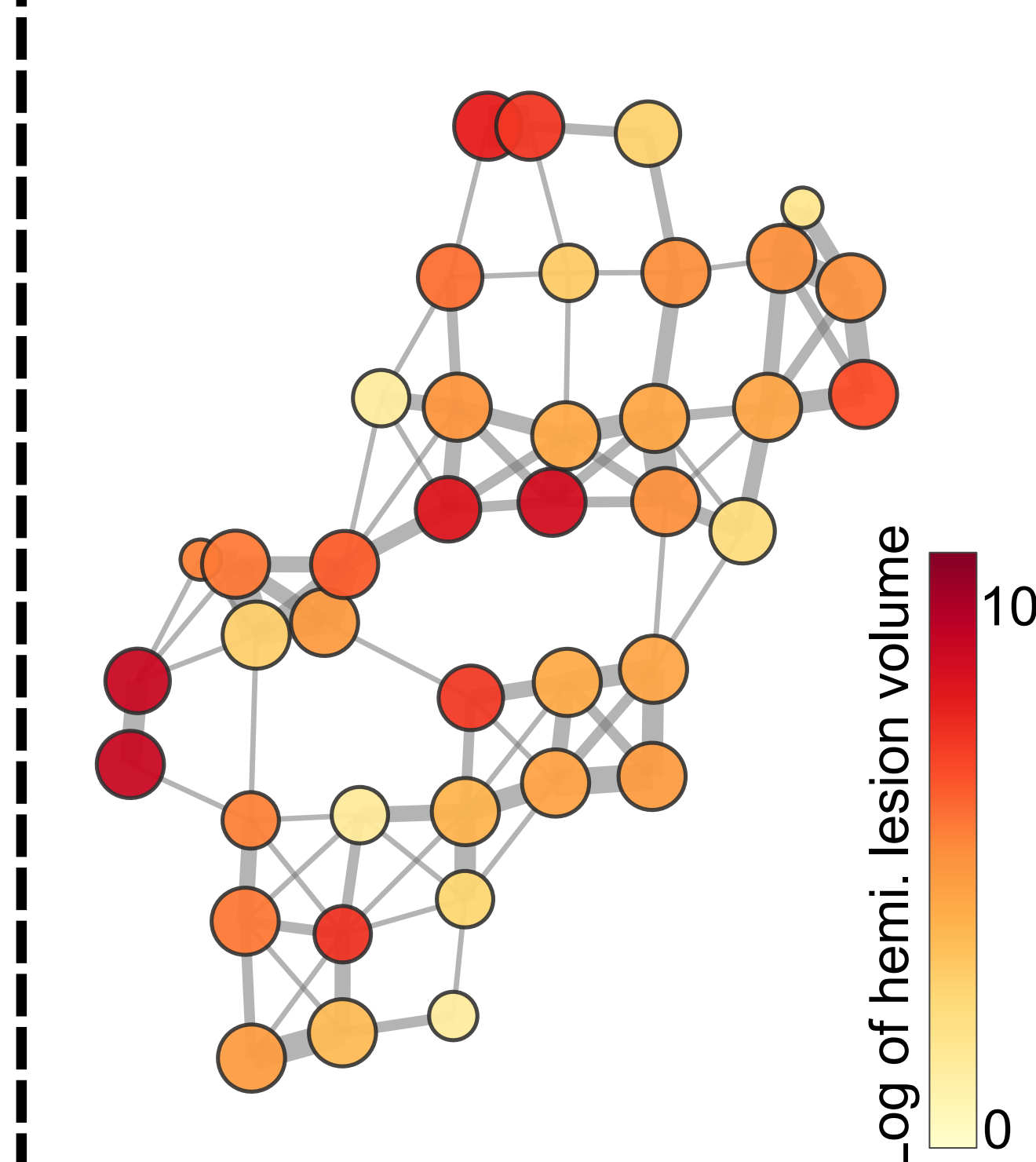


Iowa Lesion Registry Cohort

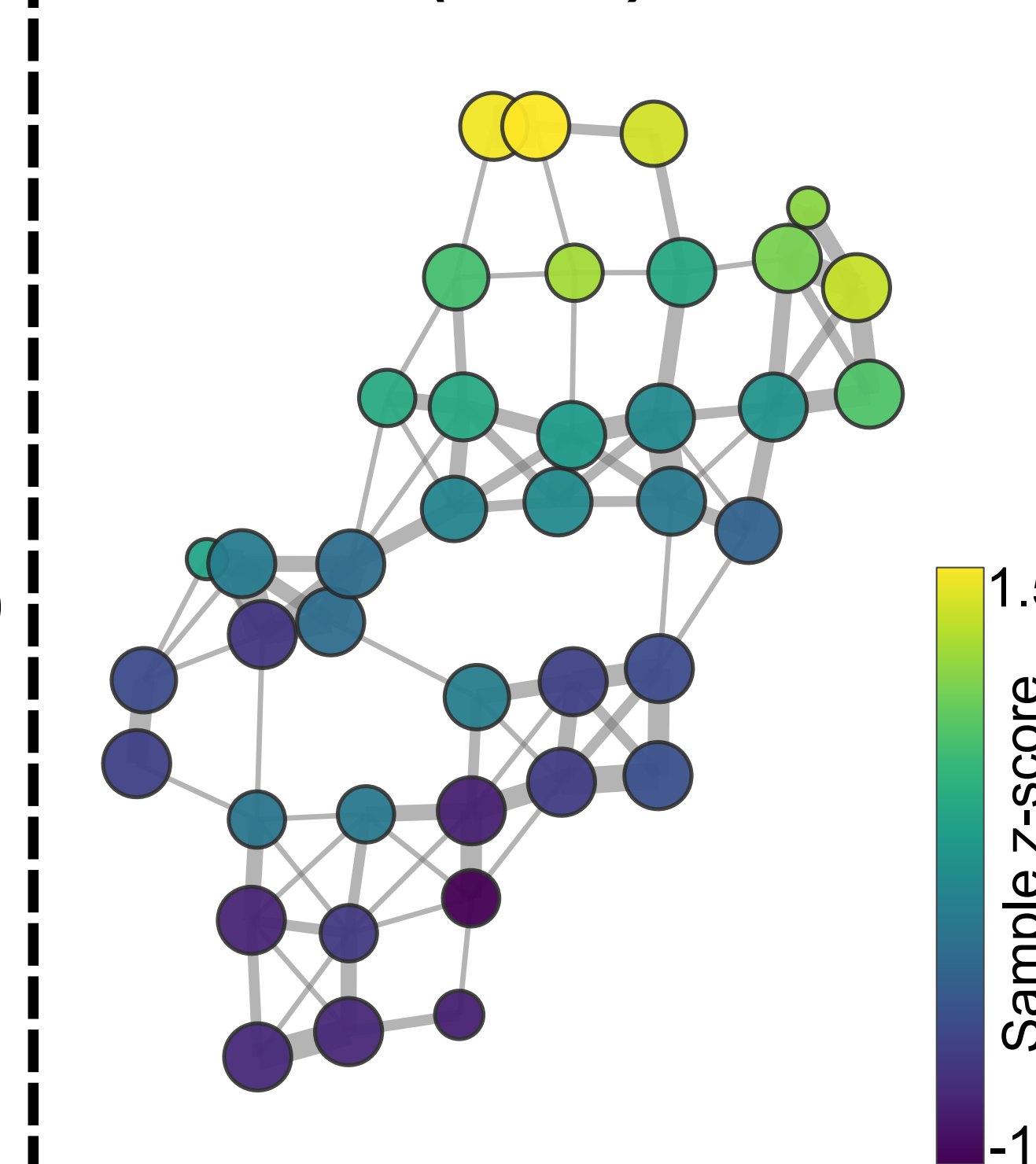
Graph constructed from chronic behavior



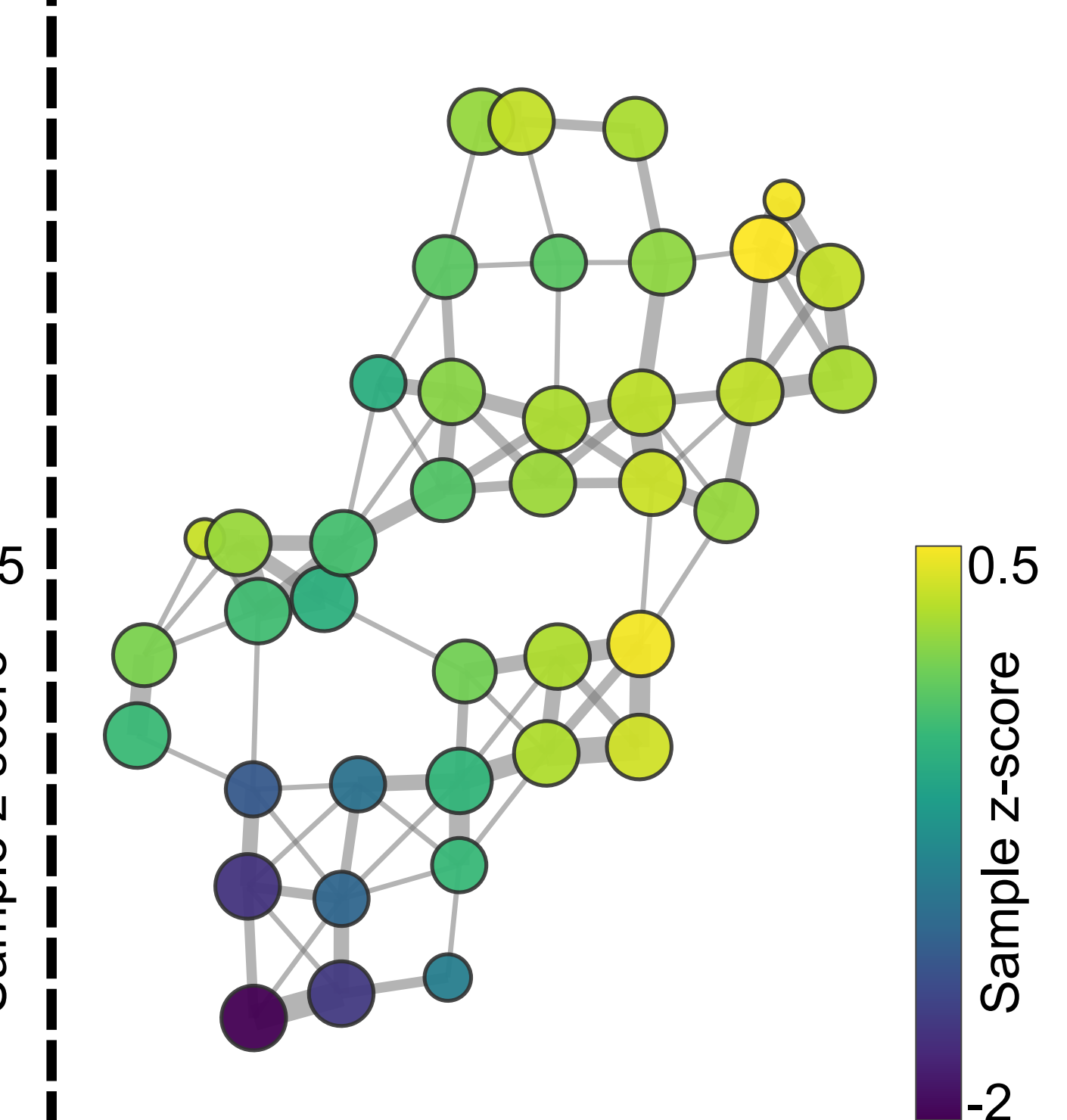
Left Hemi. Lesion Volume



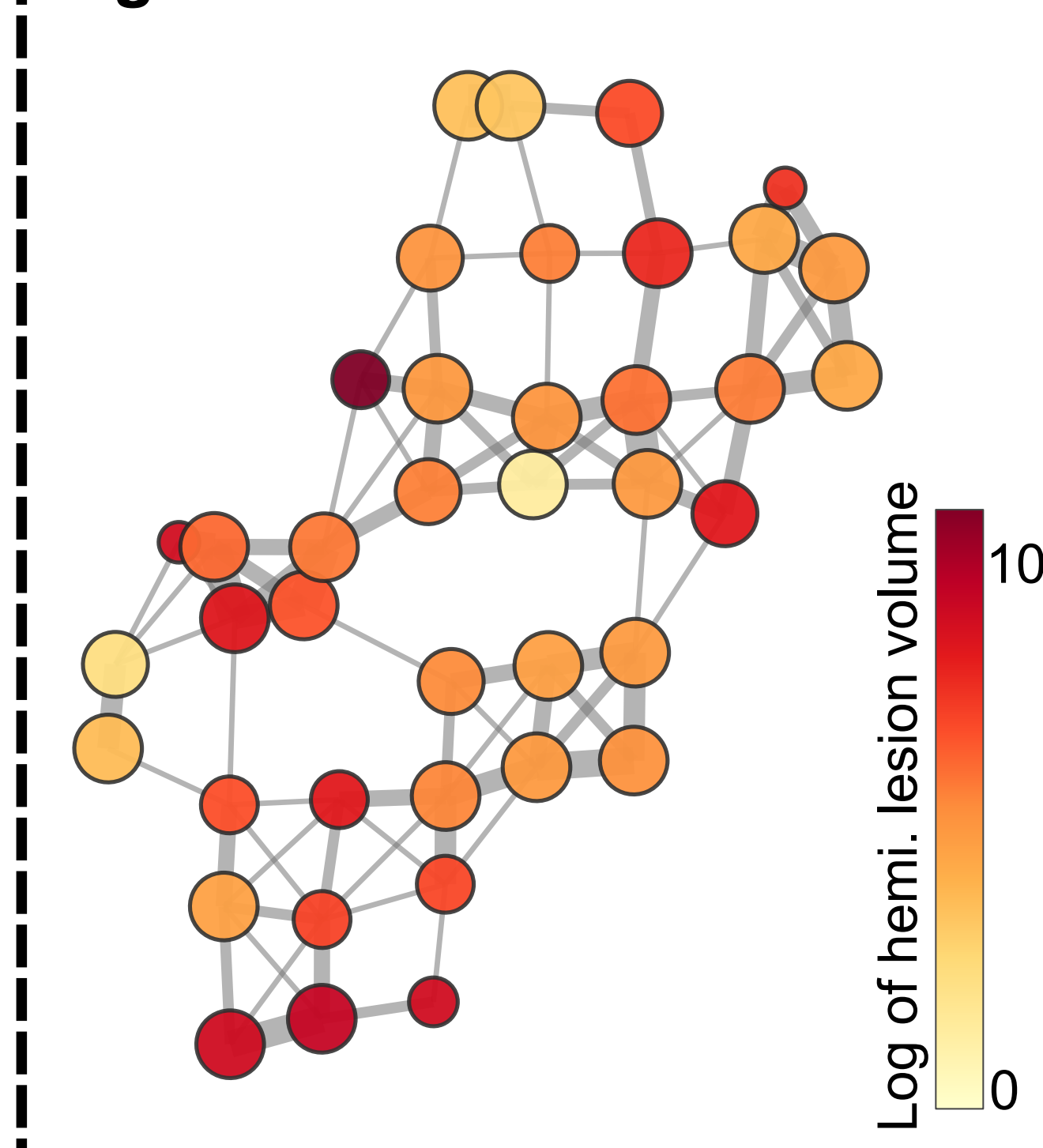
Arithmetic (WAIS)



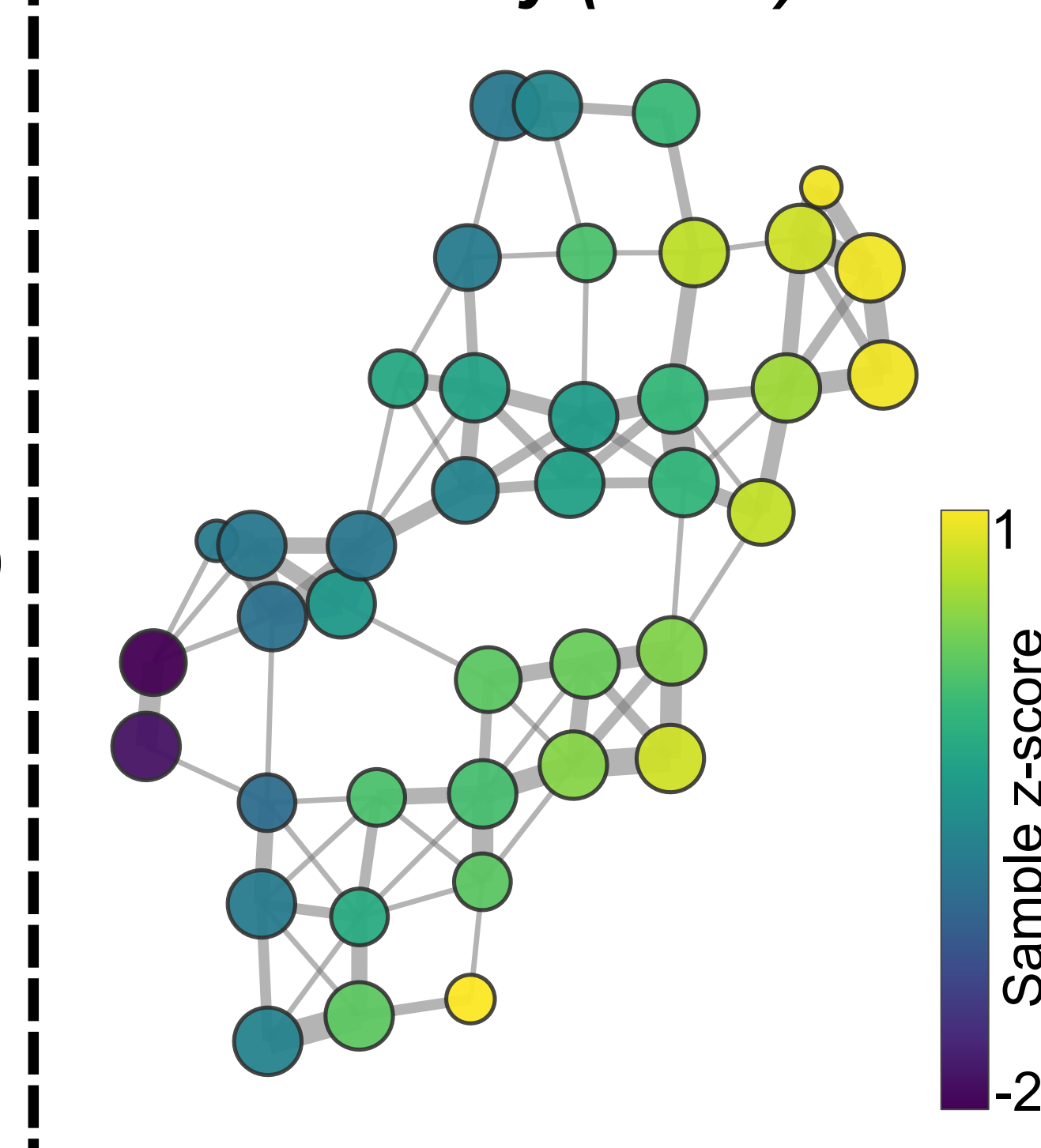
Visuoconstruction



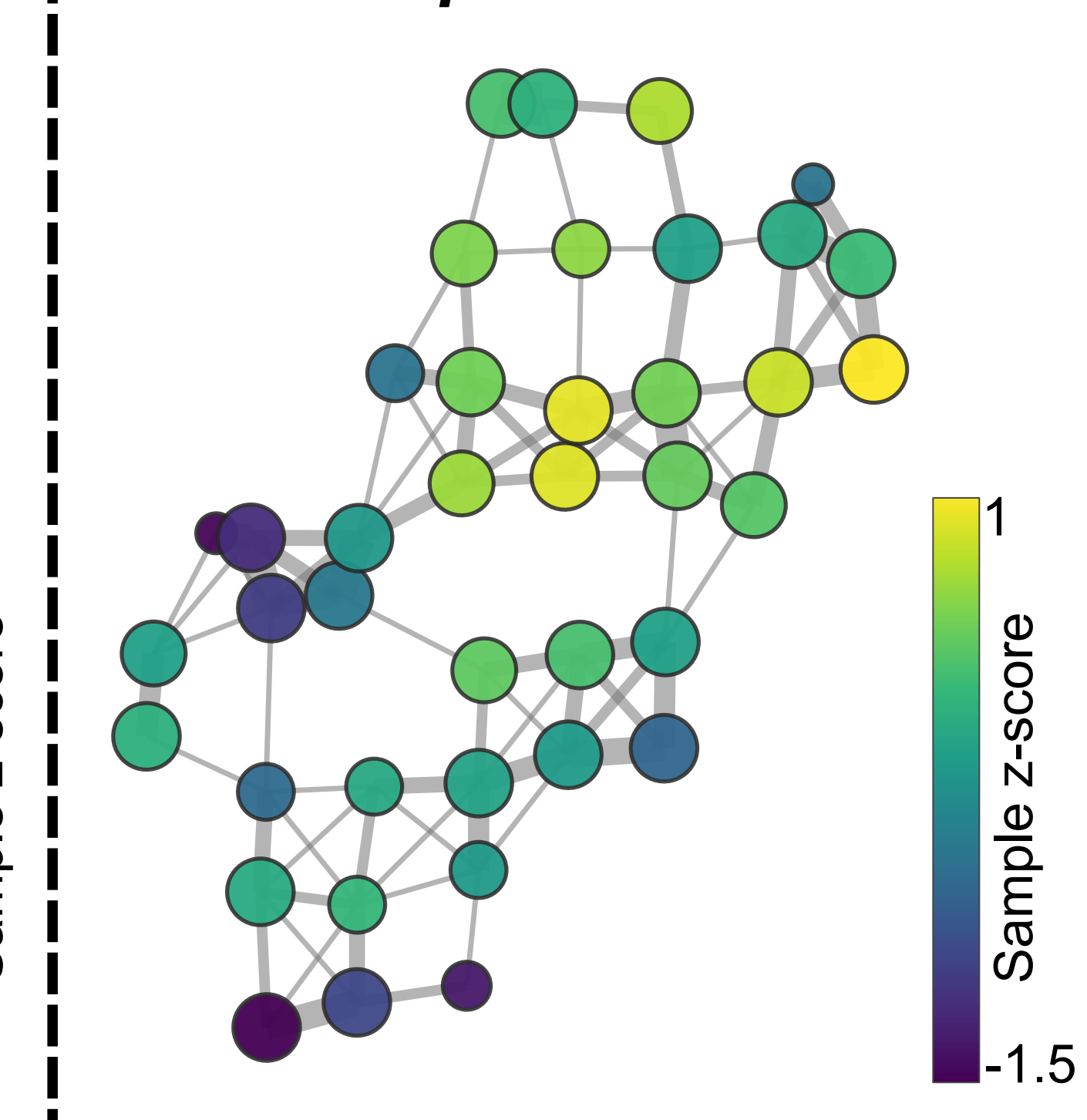
Right Hemi. Lesion Volume



Verbal Memory (AVLT)



Face Perception



Conclusions

TDA-Mapper structure:

- Reveals highly structured distribution of patients in high-dimensional space of behaviors
- Predicts multiple chronic outcomes better than standard approaches (suggestive of nonlinear relationships between acute behaviors and recovery profiles)
- Demonstrates many behaviors strongly correlate with lesion location, but some correlate more strongly with other attributes

Future Directions

- Cross-cohort prediction
- Project new patient into Mapper graph by behavior to predict lesion location and recovery
 - Explore efficacy where behavioral test batteries only partially overlap
 - Explore efficacy for imputing missing behavioral tests
- Project new patient into Mapper graph by lesion location to predict behavior and recovery
- Better characterize homologies between measured behavior, demographics, neuroanatomy (all high-dimensional, highly variable, and richly structured)
- Assess statistical significance of lesion location variation across nodes

References

- Verleysen, M., et al. (2005). *International work-conference on artificial neural networks*, 758-770.
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- Saggar, M., et al. (2018). *Nature Communications*, 9, 1399.
- Corbetta, M., et al. (2015). *Neuron*, 85(5), 927-941.